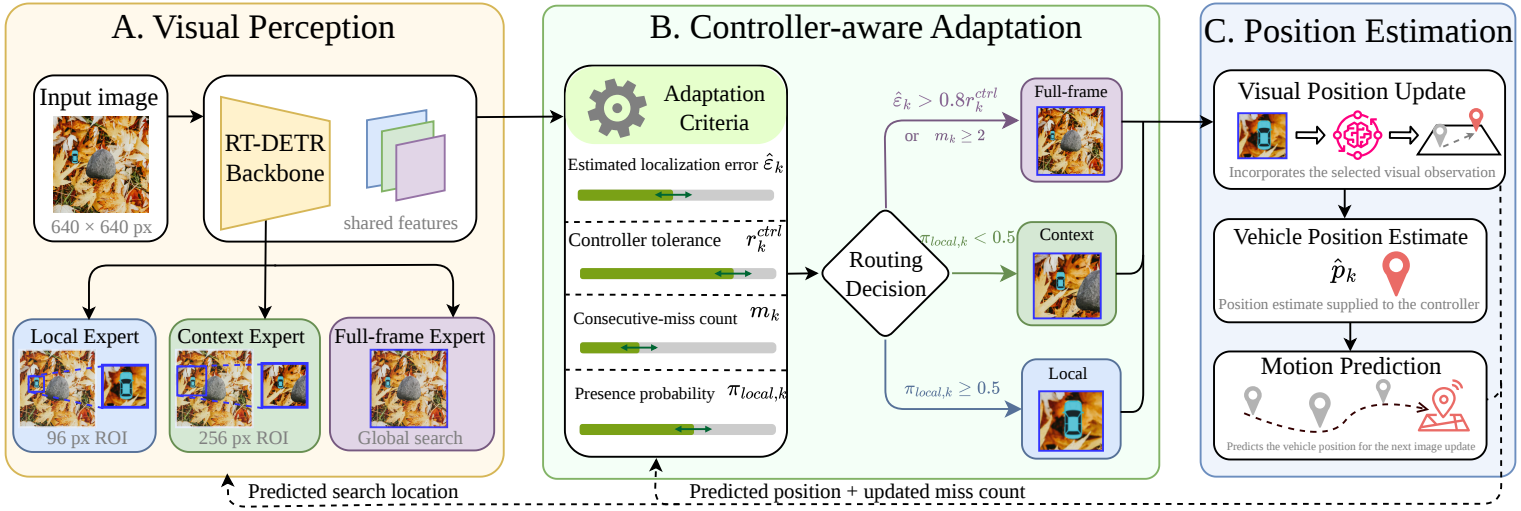

A. Visual Perception
Input image
640 × 640 px
RT-DETR Backbone
shared features
Local Expert
96 px ROI
Context Expert
256 px ROI
Full-frame Expert
Global search
B. Controller-aware Adaptation
Adaptation Criteria
Estimated localization error $\hat{\varepsilon}_k$
Controller tolerance r_k^{ctrl}
Consecutive-miss count m_k
Presence probability $\pi_{local,k}$
Routing Decision
$\hat{\varepsilon}_k > 0.8 r_k^{ctrl}$
or $m_k \geq 2$
Full-frame
$\pi_{local,k} < 0.5$
Context
$\pi_{local,k} \geq 0.5$
Local
C. Position Estimation
Visual Position Update
Incorporates the selected visual observation
Vehicle Position Estimate
$\hat{p}_k$
Position estimate supplied to the controller
Motion Prediction
Predicts the vehicle position for the next image update
Predicted search location
Predicted position + updated miss count